

Ad Hoc Semantic Reconciliation

What two reasoning agents can settle between themselves, measured across four settings and the cognition spectrum



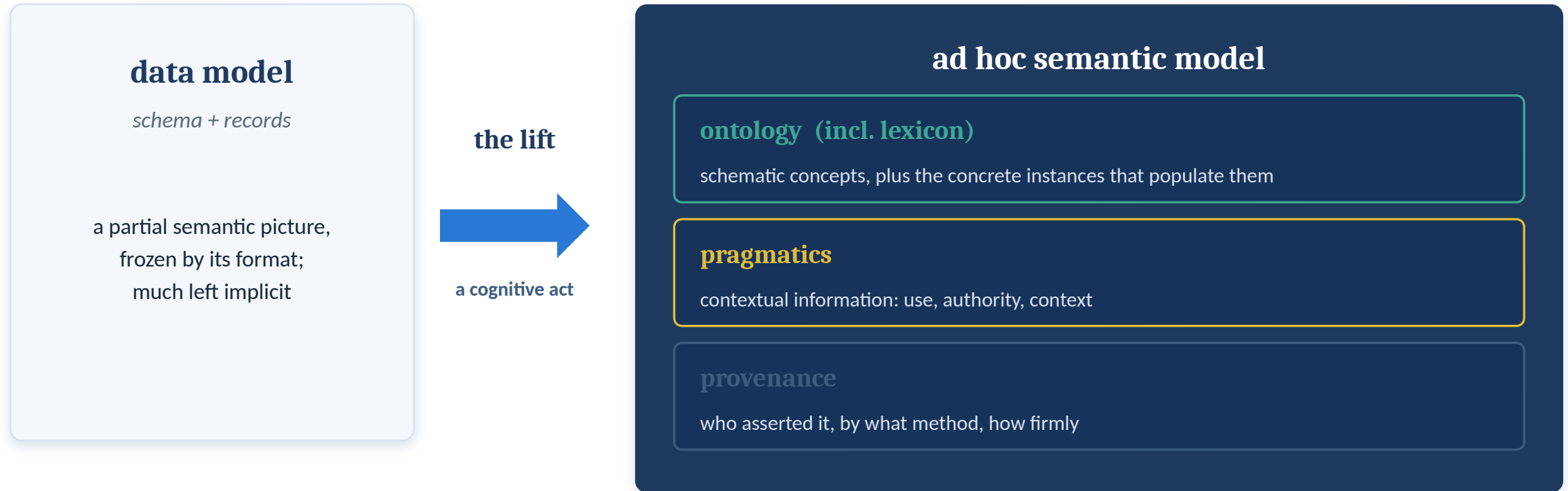
Two models, and no time to standardise

When two software systems must work together but hold different models - of the same domain, or of adjacent domains that must connect - someone has to reconcile those models. The scope a model must cover is not fixed, the useful level of abstraction depends on the task, and capable engineers, working independently, produce different but defensible models of the same network. The classical remedy is a universal standard agreed ahead of time: everyone adopts one vocabulary, and the reconciliation is done once, by committee, before the systems ever meet. That remedy is expensive, slow, and permanently behind the systems it tries to govern.

Once the systems on each side can reason, another route opens. They need not wait for a standard; they can reconcile their models ad hoc, between themselves, for the task at hand, machine to machine.

The programme puts that to the test across the cognition spectrum, from both systems reasoning to one inert to both, asking how far ad hoc reconciliation reaches at each point and what a small published reference adds. The answer is built in a measured harness, scored against a validated gold standard, rather than shown once by hand.

The lift: from a data model to a portable semantic model



A data model is already a partial semantic picture, but a fixed one, frozen by the format it uses, with much of the meaning left implicit in specifications and convention. The lift turns that frozen, partial picture into an explicit, self-describing semantic model, so any cognitive consumer can pick it up with no pre-agreed standard. Definitions, worked examples and an optional reference support the lift; they are not parts of the model.

A small family of operations

Reconciliation is not one act. Naming its operations once lets every later finding be located on a precise step of the process.

- **Lift** - data model to semantic model, per side

- **Schema binding** - align the concepts: lexical (names) and ontological (kinds, relations)

- **Instance co-reference** - decide which individual is which, sometimes by probing the live system

- **Pragmatic resolution** - settle what the reconciled thing is for, and whose authority governs it

- **Lifecycle recurrence** - re-reconcile as the live situation changes over time

- **Reference construction** - build the thin shared ground the sides bind through

- **Attribute pinning** - fix a shared field's meaning: a committed rate, not a line rate

- **Verification** - confirm a correspondence, or check that a realisation satisfies a bound

- **Composition / correlation** - group reconciled parts into wholes: several symptoms into one incident

- **Scaling** - how the total work grows with the number of systems

Where the cognition sits governs everything

Both cognitive



Reconciliation completes

Both sides reason and can be interrogated: unbounded questioning and decisive virtual experiment, so the residual is, in principle, none. This is the automatable end.

One inert



A reference buys it back

One side is a static description. Interrogation falls away; a thin published reference supplies the facts it can no longer get, up to the point of authority.

Both inert



Honest referral

Neither side can be questioned. What cannot be verified is deferred to a person rather than guessed; the residual is larger, at perfect precision.

One model ladder, one harness

THE MODEL LADDER

Three points spanning capability

 **gpt-5.6-sol**
sol · strong

 **gpt-5-mini**
mini · mid

 **gpt-5-nano**
nano · weak

Chosen so the ladder isolates model strength, not provider or architecture.

THE HARNESS

A case, scored against a gold

- A case is two lifted semantic models plus a validated gold — internally consistent, and corroborated by an independent grader: correct correspondences, known false cognates, and the residual expected at each placement.
- Four factors are varied: the reference (none, thin, richer), the cognition placement, the operational case, and difficulty.
- Five metrics are read: precision, resolved fraction, surviving false cognates, the residual, and reasoning tokens.

One thesis, tested four ways

1 • Configuration

ONF TAPI ↔ IETF TE / ACTN

Two public standard models of one network. The founding test: cognition reconciles them ad hoc, and a reference substitutes for it.

2 • Intent

refine • negotiate • lifecycle

A declarative intent reconciled against a concrete service by refinement. A two-sided negotiation, and the pragmatic operation enters.

3 • Cross-domain

private OSS ↔ private IP/VPN

No public standard beneath either model. The agents must build the shared ground themselves before they can bind.

4 • Observability

legacy fault mgr ↔ IETF NMOP

An alarm is not an anomaly. Meaning is settled by a standard; the whole question is significance, carried by pragmatics.

Two standard models of one network

THE CASE

ONF TAPI and IETF TEAS/ACTN describe the same optical network in different ways, each in its own vocabulary and structure. A TAPI connectivity-service is a TEAS tunnel; a TAPI link-termination-point looks like a TEAS tunnel-termination-point but is a link end, not a trail head. Both are public standards the agents recognise, which makes this the cleanest place to isolate the mechanism.

WHAT WAS PROVEN

Cognitive agents reconcile the two models correctly and ad hoc; the concept holds. Progress shows in two numbers: precision (of the correspondences an agent commits, the share that are right) and resolved fraction (of the true correspondences, the share it confirms rather than defers). The lift is the lever: matching on bare labels finds only 0.66 to 0.76 of the correspondences, and the lifted content recovers them to 0.92 to 0.97. Verification, run as its own step, rejects the false cognates, so precision climbs toward one. A thin reference then carries the strong agent to a perfect, verified close, both numbers at 1.00 across the whole spectrum, for one to two orders less deliberation. For the weaker agents the reference mainly prevents the errors they would otherwise commit, lifting hard-case precision (mini 0.88 to 0.96, nano 0.91 to 0.97) short of that perfect close. Work scales linearly with a shared reference and quadratically without, verified to $N = 12$.

Cognition completes the reconciliation; a reference partly substitutes for it, saving cognitive effort and preventing errors, and the spectrum governs both. Setting 1 does not touch pragmatics at all; that is the thread the other three settings pick up.

Refinement, negotiation, and a self-renegotiating service

THE CASE

A customer's intent (New York to Frankfurt, at least 8 Gbit/s, at most 5 ms, four-nines) is a set of bounds, not a service. An operator holds a catalogue of concrete optical services. Neither speaks the other's language; they must work out which realisation satisfies the wish and negotiate when none meets every bound.

WHAT WAS PROVEN

With both agents live, the reconciliation completes autonomously, negotiation included, at decision accuracy 1.0 for sol and mini, with no human in the loop. Remove the customer's live judgement and a pre-placed movable policy still closes the decisions it was authorised for (1.0); a mute customer with no policy cannot, and the agents correctly refer the decisions onward rather than guess. A published reference supplies information: the blind strong agent rises from 0.29 to 0.71 once given an invariant floor to check against. It does not supply authority: at both-inert no reference moves the negotiation, because what is missing is the customer's judgement. Effort climbs steeply down the ladder, from about 150 reasoning tokens for sol to 860 for mini and 5,400 for nano.

Cognition completes the refinement. A thin published reference can stand in for the information an inert side would otherwise supply, but not for the authority to decide, and a pre-placed policy is shown able to carry a party's authority to where a person or live cognitive system would otherwise have to stand — a control with an actual second reasoning agent in the provider's place changes the outcome almost nothing where the provider is a fixed catalogue of offers; give the provider its own deliverability to weigh and the second agent diverges substantially, so this is a negotiation that is two-sided where the provider has something of its own to reason about.

Reconciling with no public standard

THE CASE

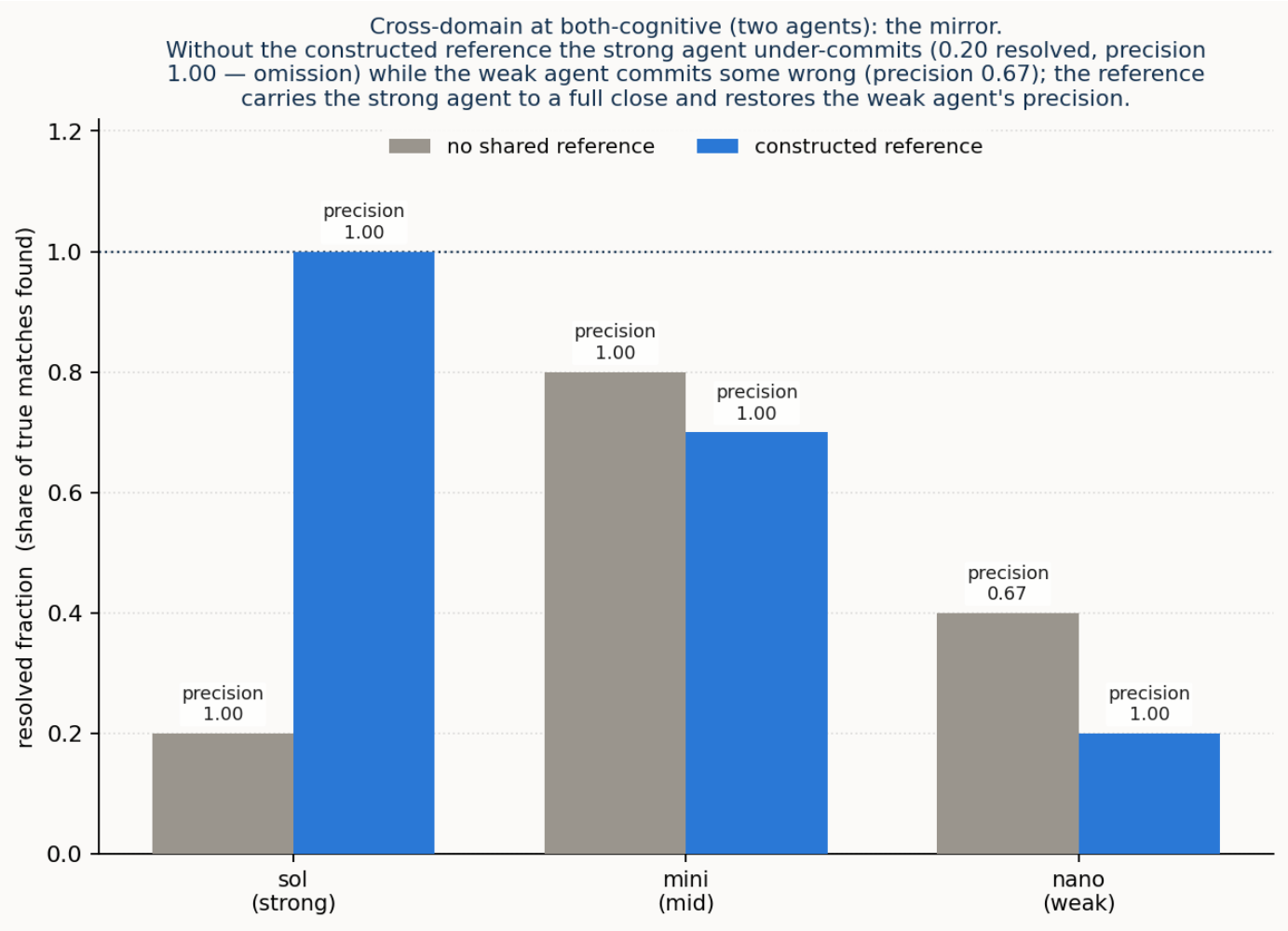
Meridian, a private transport OSS, thinks in circuits, bearers, wavelengths and protection grades. Cascade, a private IP/VPN controller, thinks in services, attachments, VLANs and classes of service. Their worlds touch at one seam. Both carry a grade, but Meridian's is a transport protection class and Cascade's an IP class of service: the same word, unrelated meanings, and no standard to consult.

WHAT WAS PROVEN

The central result is a pair of mirror-image failures. Without a shared reference the strong agent under-commits: it binds only the names that already coincide, resolving about a fifth of the correspondences (resolved fraction ~ 0.20) at perfect precision, and defers the rest. The weak agent's opposite mistake is commission, and it shows where a single agent reconstructs a side on its own: its precision never clears 0.83 and it takes the grade trap. When the two agents negotiate, each proposal must be ratified by the other, so the weak agent cannot bind alone and the trap does not survive. A thin reference remedies both, and the agents need not be handed it: building the shared ground themselves lifts the strong agent from 0.20 to 0.80, and to 0.90 with a decisive experiment, near the reference-given close, at perfect precision. The reference works through shared description, never through a bare pointer.

With no public standard, cognition still closes, but only once it has built the shared ground, and building that ground is the work. The reference reaches meaning; it does not reach authority.

The mirror: the strong agent omits, the weak commits



Precision against resolved fraction, reference off (hollow) to on (filled). Without the constructed reference the strong agent commits little and all of it is right, while the weak agent commits much of it wrongly; the reference pulls both to the corner.

An alarm is not an anomaly

THE CASE

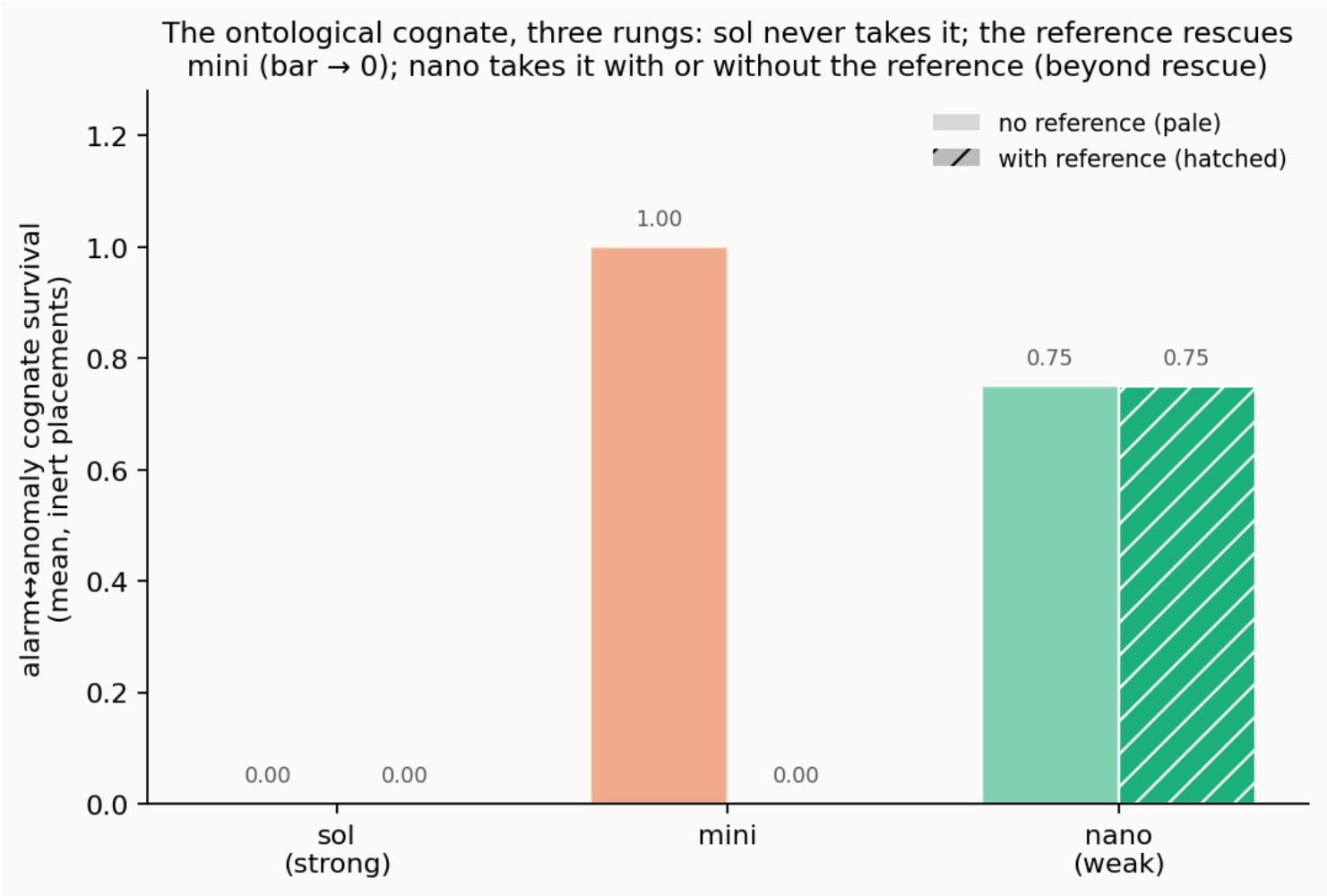
An alarm and an anomaly look alike but are not the same thing — a false cognate. An alarm is a device's own notification; an anomaly is a cross-signal judgement that something is wrong.

WHAT WAS PROVEN

Telling them apart is a three-rung capability gradient — the strong agent never conflates them, with or without the reference; the reference rescues the mid agent completely (the cognate's survival goes $1.00 \rightarrow 0.00$); the weak agent conflates them either way.

It is the programme's deepest false cognate, and only cognition — or a reference for the middle of the ladder — tells the two apart.

The alarm-anomaly cognate is a three-rung gradient



Survival of the alarm-versus-anomaly false cognate at the inert placements, without and with the reference. The strong agent never takes it; the reference drives the mid agent to zero; the weak agent barely moves.

Ad hoc reconciliation works

Two cognitive agents reconcile independently authored, divergent models correctly, with no standard agreed in advance. They complete an equivalence between two standards, negotiate an intent, bind across a private domain boundary, and judge what an observability signal means. Every case is scored against a validated gold standard across the cognition spectrum.

At the fully-cognitive end, the reconciliation completes autonomously in every setting: no standard agreed in advance, no human in the loop.

This holds even for the two operations that look least automatable: **a two-sided negotiation (decision accuracy 1.0) and a significance verdict (accuracy 1.0 with the pragmatics on).**

Everything else in the synthesis qualifies this: how automation degrades as cognition recedes, and what a thin reference buys back.

Full cognition makes a reconciliation self-verifying

Placement on the cognition spectrum decides what a reconciliation can bind. It also decides something that cuts across all four settings: whether the agents can verify their own result, confirming the binding is right rather than merely asserting it.

With full cognition on both sides, the same mechanisms that close the residual, mutual interrogation, bilateral ratification, and a decisive virtual experiment, also confirm the close. The agents verify their own result rather than assert it.

As a side goes inert those mechanisms fall away, verification degrades to a weaker check by satisfaction, and the assurance that the reconciliation is correct weakens with it. Full cognition does more than improve accuracy: it is what makes an ad hoc reconciliation safe to automate, with no standard and no person in the loop.

How two agents complete a reconciliation between themselves

The negotiation

Each agent holds only its own model and a short list of the other's concepts, and neither can bind on its own.

They take turns: advertise a concept, ask and answer questions about it, then propose a match. It counts only when the concept's owner accepts it. That mutual check is what holds precision high and refuses the look-alikes, with no standard in play.

Run across the whole capability ladder, this check keeps precision high and drives false matches to zero for every model but the weakest. The structure itself is doing the verifying.

The decisive experiment

For a match they are still unsure of, the agents can settle it by trying it, not just discussing it.

One agent sets the pair up in a private virtual copy of the network, runs it, and reads back whether the result holds. The verdict goes to both sides and settles the pair: confirm binds it, refute drops it. This is how two agents reach a full close on their own, or reject a look-alike, with no reference to consult.

So the fully-cognitive end completes and checks itself. That, not accuracy alone, is what makes it safe to automate with no standard and no person in the loop.

What the four settings establish

- T1** It is cognition that completes a reconciliation: descriptor methods carry it most of the way, then reach a ceiling and stop.
- T2** Where the cognition sits is the master factor: the further it recedes from both agents, the more the reconciliation leaves unresolved.
- T3** A thin reference stands in for some of the reasoning and supplies facts an inert side lacks, but never provides the authority to decide.
- T4** The two kinds of failure are opposite: a strong agent leaves correspondences out, a weak agent binds them wrongly. When two agents negotiate, each must ratify the other's proposals, and that check removes the weak agent's wrong bindings across the whole capability ladder.
- T5** The pragmatic layer, what a thing means in context and whether it matters there, lies beyond names and references; it is decisive and gated by the agent's power.
- T6** A shared reference makes the work scale linearly rather than quadratically with the number of systems; which fields it carries also matters.

The four evidence readings that follow take these along four cross-cutting axes: cognitive load, the reference, placement and failure, and the pragmatic layer.

Capability buys economy as much as correctness

To reach the same decision, the weak agent spends one to two orders of magnitude more reasoning than the strong one. Effort was metered directly in the two settings that turn on a discrete decision, the intent negotiation and the observability verdict, where the reasoning to reach it can be counted; the pattern holds in both. A weak agent does not merely make more mistakes; it burns far more cognition making them.

Intent negotiation

reasoning tokens to a decision

sol ~150

mini ~860

nano ~5,400

about 36× sol to nano

Observability verdict

reasoning tokens to a decision

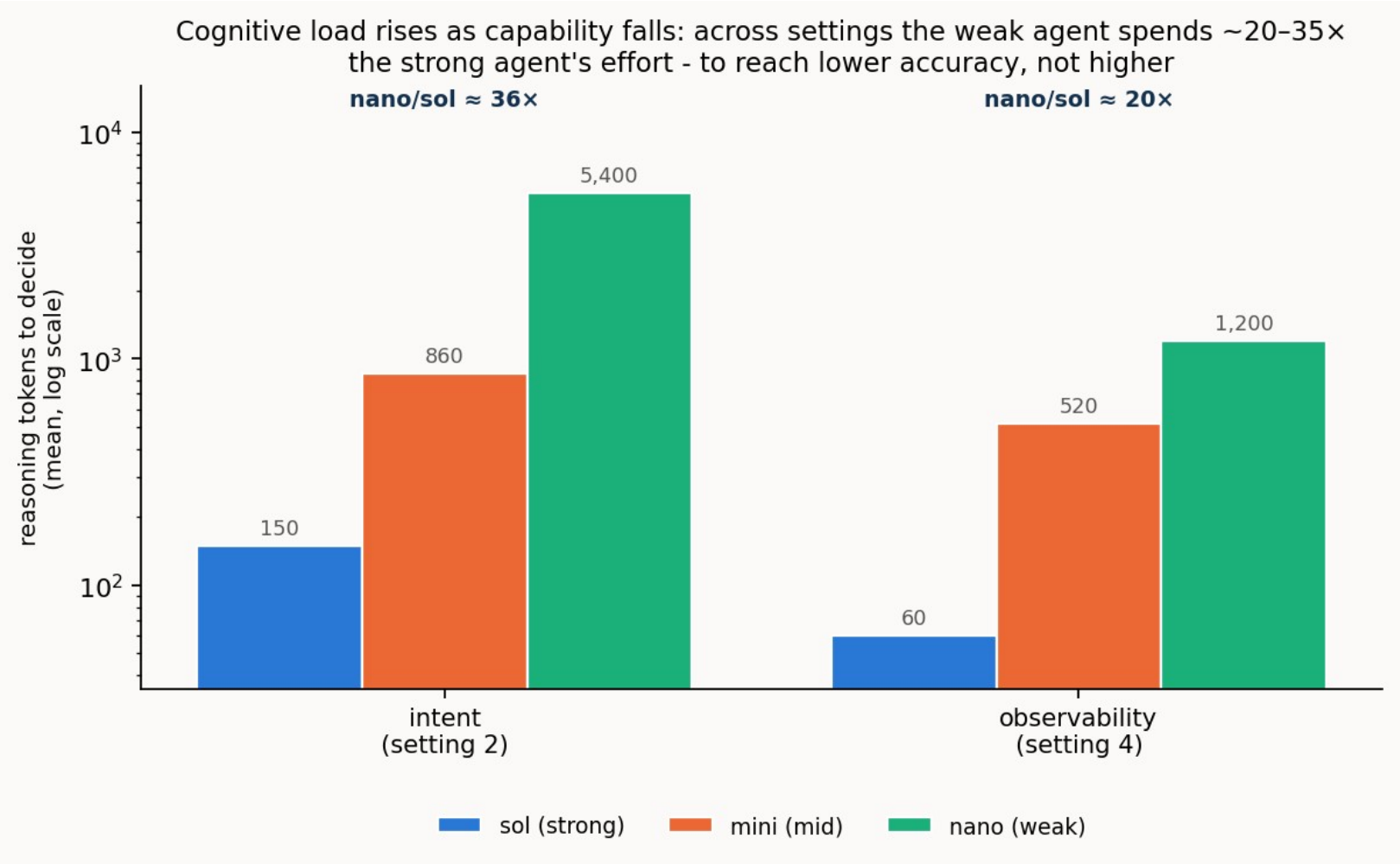
sol ~60

mini ~520

nano ~1,200

about 20× sol to nano

A weak agent spends far more to reach less



Reasoning tokens to reach a decision, per model, in the two settings that measure effort (log scale). The weak agent spends roughly 20 to 35 times the strong agent's effort, and to reach lower accuracy.

One thin reference, a different benefit at each end of the cognition spectrum

CORRECTNESS VALUE

Concentrates where cognition is weak

The weak agent is the one that commits false cognates; the reference disciplines it out of them. A strong agent was already going to get those right, so there is little correctness left to add.

EFFORT VALUE

Peaks where cognition is strong

A capable agent that would otherwise reason its own way to the answer leans on the reference instead, cutting its effort by one to two orders. In the configuration setting this collapse in the strong agent's deliberation, with the reconciliation still perfect, is the reference's main payoff.

...but its potential fields are not of equal value

Even one descriptive field (a shared label, a definition or an example) lets a capable agent complete the binding. A bare shared identifier with no description is worse than nothing (precision 0.50); a shallow class tag can even mislead the weak agent it was meant to help. A reference works through shared meaning, never through the pointer.

The further cognition recedes, the more is left unresolved

Probing resolves the hard cases until a side goes inert

instance co-reference • setting 1

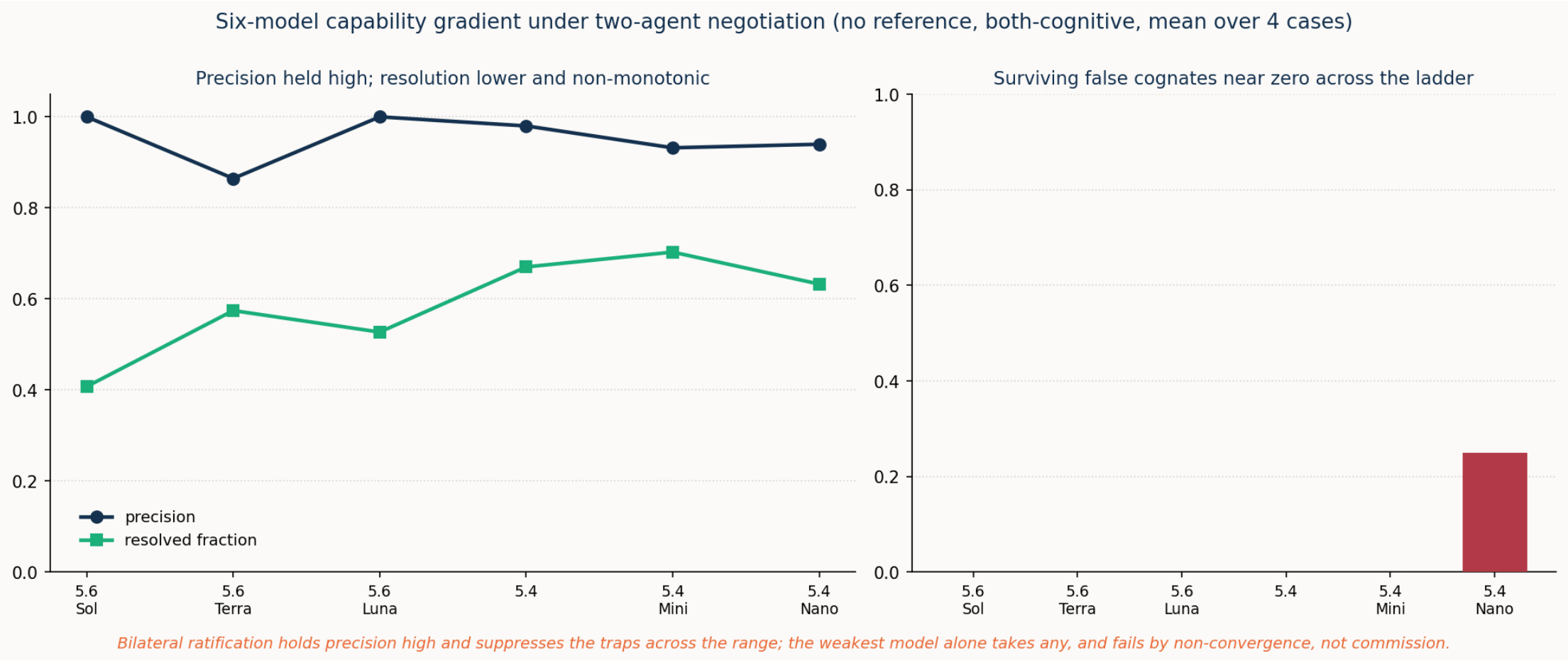
- At both-cognitive, a live side can be interrogated: more probing drives the residual to zero, so the only limit is how much you probe. This is the budget-limited case.
- At one-inert, there is nothing live to probe, so no amount of probing helps and the residual sticks near 0.08. The limit is now structural.
- The same probing is run at both placements; whether it resolves the cases is set by placement, not by how much effort is spent.

Strong and weak agents fail in opposite ways

the strong omit • the weak commit

- Denied the ground it needs, a strong agent defers: it leaves the hard cases unresolved, at perfect precision on what it does bind.
- A weak agent's opposite mistake is commission: binding wrongly. That shows where a single agent reconstructs a side alone; when two negotiate, each proposal is ratified by the other and the wrong bindings are refused.
- In the cross-domain setting the two break in opposite ways on the very same case; and swept across six models, negotiation compresses the gradient, holding precision high and suppressing the false cognates the whole way down.

Two-agent negotiation compresses the gradient



Precision, resolved fraction, and surviving false cognates across a six-model capability ladder (two-agent negotiation, both-cognitive). Bilateral ratification holds precision high and drives false cognates to zero across almost the whole ladder; resolution is lower and non-monotonic.

The dangerous failure is the confident wrong answer

CONFIDENT-ERROR RATE, DOWN THE LADDER

0.004 → **0.031** → **0.046**

strong → mid → weak — about 11×, and higher again as a side goes inert.

CONFIDENCE WHEN WRONG

0.73 – 0.85

only ~0.15 below its confidence when right — too thin to filter on.

Confidence is a real but weak guard

A consumer cannot separate right merges from wrong ones by the number. The remedy is an explicit abstain path — a scored “insufficient evidence” outcome, reported apart from honest deferral — not a confidence threshold.

Decisive for meaning and gated by cognitive power

What a reconciled thing is for, whether it matters in its context and what to do about it, lies beyond names, glosses and references. It is settled from the pragmatic information a lifted model carries: use, authority, context. The observability setting shows this by example, in two operations that resolve significance from that information:

The verdict

judgement • power-gated

Deciding whether an anomaly warrants a page needs judgement. With the pragmatics on, sol and mini suppress correctly during maintenance (1.0 and 0.83); handed the identical annotations, the weak agent barely moves (0.53 on, 0.50 off).

Correlation

structural input • robust

Folding related symptoms into one incident needs only the dependency map. Given it, every model on the ladder, the weak one included, correlates correctly; without it, every model fails.

Both draw on that pragmatic information and both are decisive; but the verdict is a judgement while correlation only applies a structural map. That difference, not the label, decides whether a weak agent can do it.

Three layers, told apart by how each is obtained

Schematic layer



the TBox • standing structure

The concepts, kinds and relations. It changes only when the model is re-lifted. The lift holds it in full, and an agent recovers it from the schema surface at every capability, even with the names stripped.

Concrete layer



the ABox • observed

The instances that populate it now. Dynamic, but obtained by observation: a re-reading of the live source that any consumer, even a non-cognitive one, can do. The lift holds it as a refreshable snapshot.

Pragmatics



the interpretation • judged

Whether a state matters, whether an offer is acceptable, who decides. Not read but judged, from schema, instances and context; defeasible and context-relative. The lift can only describe it; no static model, ours or OWL, holds it resolved.

The pragmatic layer is supplied by the live operation

1

Pull in

Interrogate a live peer. The open question, “what facts do you have?”, pulls a pragmatic fact the lift never carried, from a counterpart, a catalogue, or a resource.

2

Refresh

Re-observe the current context and re-resolve against it: the concern is low because the maintenance window is open now, the value the lift could only describe.

3

Create certainty

Run the decisive virtual experiment: provision a candidate, operate it, and read back whether the required invariants hold, making a certainty no lift contained. The same mechanism as an intent satisfaction check.

4

Resolve

Settle the pragmatic verdict itself, the significance, the acceptability, the authority, by judgement against the current context. This is the capability-gated frontier.

The cognition spectrum and what happens at each placement

Placement	What the reconciliation can do	The residual	What a reference buys
both-cognitive	completes autonomously in every setting: bind, negotiate, verdict	none, in principle	substitutes for cognition (effort down 1-2 orders)
one-inert	live side reconstructs the mute one; some resolutions turn structural	grows; the shortfall appears	supplies missing information; rescues the mid agent
both-inert	a third party can only propose for adjudication	largest; every judgement referred onward	supplies information, not authority

The three rows are the master variable; every finding in the synthesis locates on one of them.

Outside the spectrum: a pre-placed policy (cognition captured in a portable artefact ahead of time) can partly stand in for a live agent in some cases, closing the decisions it was authorised for and carrying that authority.

The operations, and where each was put under test

Operation	1 • Configuration	2 • Intent	3 • Cross-domain	4 • Observability
Lift	the lever (0.66→0.97)	reused	both sides private	alarm decomposed
Reference construction	given (lexical)	given (invariant)	constructed 0.20→0.80	given (RFC 9940)
Schema binding	reference substitutes	shared ground	omission vs commission	ontological cognate
Attribute pinning	—	bound vs measured	committed vs line rate	severity / scores
Instance co-reference	budget → structural	measured (endpoints)	bracketed (repro s.1)	measured; repro s.1
Verification	own step	by satisfaction	isolated pass	verdict stands in
Pragmatic resolution	deferred	movable policy	authority attribution	verdict, power-gated
Composition / correlation	—	—	—	robust (all models)
Lifecycle recurrence	—	four-hop loop	—	—
Scaling	linear vs quadratic	—	—	—

Five surprises worth stating as findings

1

A strong agent scoring lower is often behaving better

Blind at both-inert, it refuses to affirm what it cannot verify: honest deferral rather than weakness (setting 2: 0.29 against 0.71). Read the accuracy without the precision beside it and you misjudge it.

2

A bare shared pointer is worse than no reference

An opaque identifier with no description drops precision below the no-reference floor. The reference works through shared description, not the token; even a shallow class tag can mislead.

3

What lets a weak agent resolve more also gives it more to get wrong

The same lifted content that helps it find correspondences also hands it more material to misfire on. The lift is unambiguously safe for the strong agent and double-edged for the weak.

4

Two operations in one setting demand opposite things

In observability the verdict is a judgement while correlation only applies a structural map. Both resolve significance from the same pragmatic information, but only the demand for judgement predicts which a weak agent can do.

5

A “capability-independent” law was one model family's habit

The record-reading verifier's ~0.71 ceiling looked structural; across three families it splits 0.71 / 0.76 / 0.84 and capability does not order it. Only replicating across families told the disposition from the law.

How far an agent reaches, and that the findings replicate

Reach: what an agent can see and ask

Hold the strong agent at its best and vary only what it may SEE (how much of a record) and what it may ASK of the live system.

Seeing more caps the hardest cases at one-half. Being able to ask the OPEN question - '*what facts do you have?*', *not a named field* - takes it to a full close, at perfect precision. The decisive reach is the question you did not know in advance to ask.

Across the ladder the same access lands by capability: the strong agent converts it to a full close; the mid plateaus; the weak posts a high resolved fraction but with slipping precision. Power sets the ceiling reach can reach.

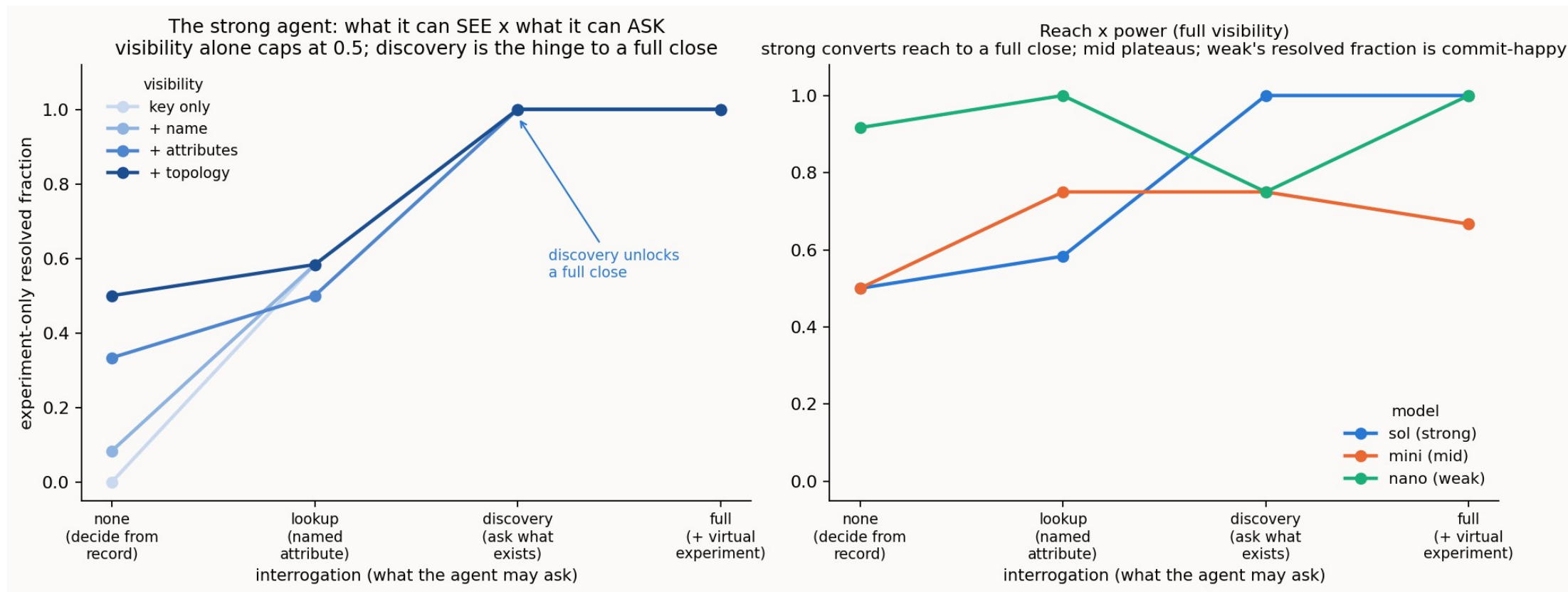
Breadth: the findings replicate

Each setting is now exercised on THREE independently-built cases, not one - different domains, vocabularies, traps.

Every signature reappears: the strong agent closes and the reference prevents the weak agent's errors (configuration); the omit-vs-commit mirror, repaired by a reference (cross-domain); refinement and negotiation complete while the lifecycle grades with capability (intent); the alarm-is-not-an-anomaly look-alike reliably avoided (observability).

It closes the single-constructed-case worry as far as in-house work can - but not real network data (same hand built these), which is what remains.

The decisive reach is the question you can ask



How far the strong agent settles the hardest cases as what it can see and what it can ask are varied (left), and the same across the model ladder (right). Seeing more caps at one-half; being able to ask the open question takes it to a full, correct close.

The findings replicate across model families — and one “law” did not

WHAT TRANSFERRED

Re-run against two other families and tiers — an open-weights Qwen-2.5-7B and DeepSeek at the strong tier, through the same client — most findings hold: contradictory authority sources are surfaced by every model, and the external-artefact remedy rescues every family and tier (0.47 → 0.93; 0.40 → 1.00).

WHAT IT CORRECTED

The record-reading verifier's pass rate near 0.71 had looked capability-independent. Across families it is not a ceiling but a disposition — 0.71 / 0.76 / 0.84 for three families — and capability does not order them. What read as a structural law was one family's habit.

Only replication tells a disposition from a law

Replication supplies the rigour a single family cannot: an independent grader from a second family corroborates the derived gold, rejecting the same trap and cross-framing pairs.

The harder half — producing the inputs

ALIGN

Two independently authored lexicons, no shared id, align on meaning: the capable agent holds precision 1.00 — the harder half of the schema bind.

REUSE

Onboarding a genuinely new system against a shared reference is linear, not quadratic: 2.0–2.7× cheaper in tokens and more accurate at once.

DERIVE THE MAP

The dependency map correlation runs on is derivable from raw inventory; correlation on the derived map matches the given-map baseline.

DERIVE AUTHORITY

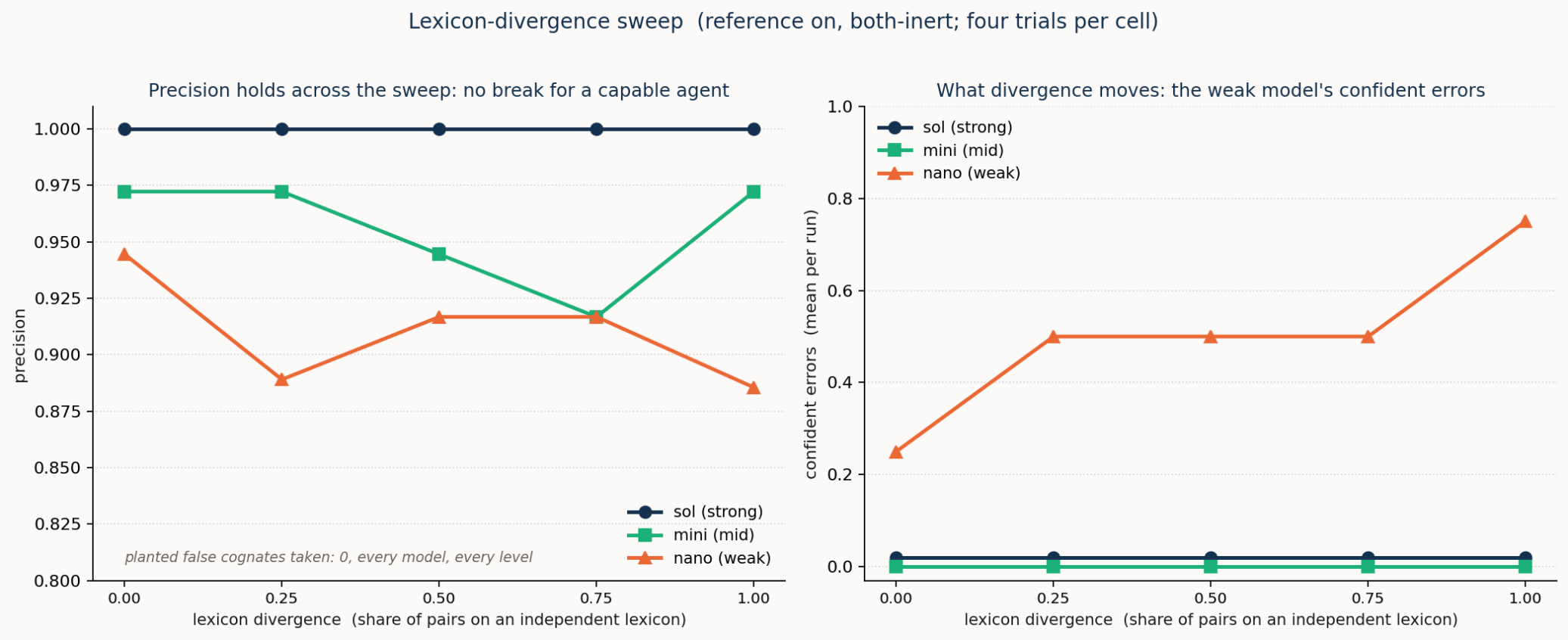
Cross-domain authority is derivable from an external artefact (0.47 / 0.40 → 1.00); contradictory sources are surfaced, not silently resolved.

ROBUST TO DIVERGENCE

Swept from a shared reference to two fully independent lexicons, a capable agent's alignment holds precision 1.00 throughout — no break found.

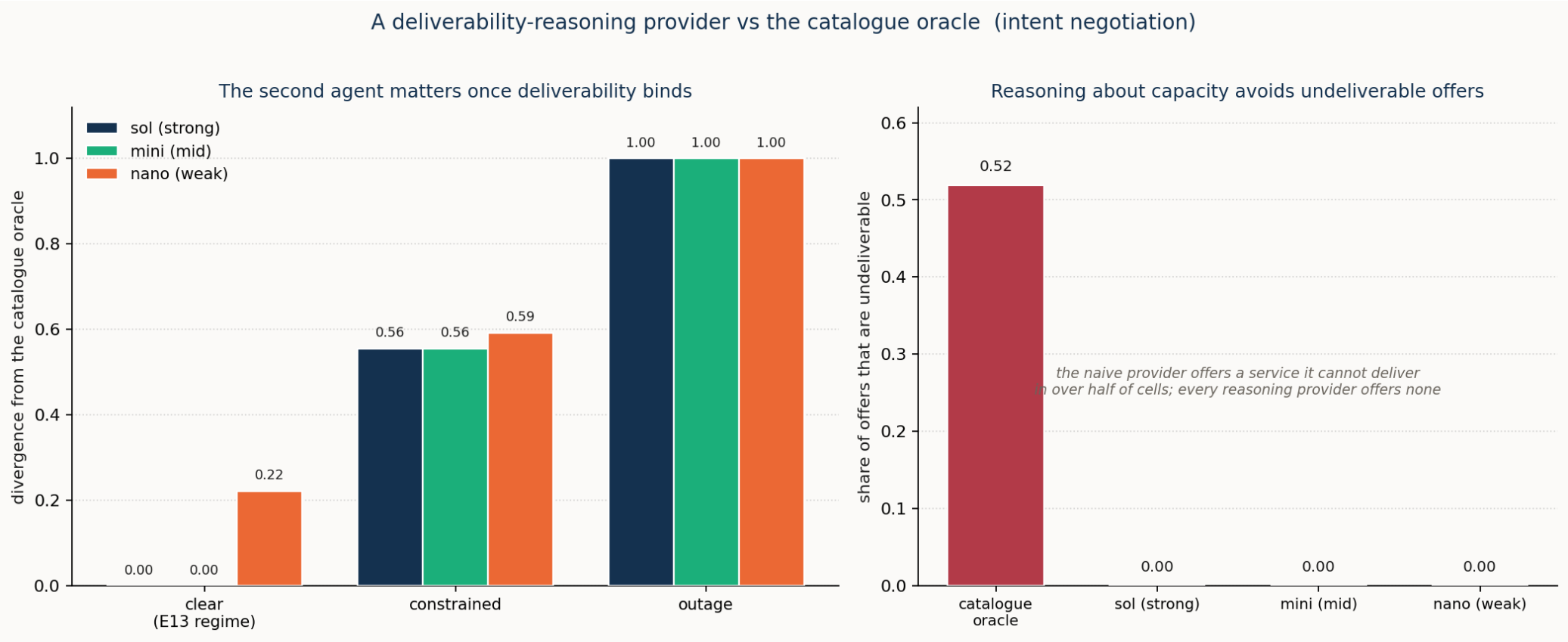
Producing the inputs is often achievable; the external artefact or reference is the reliable route; the residual risk is the capability-gated confident error.

No break for a capable agent



Sweeping from a shared reference to two fully independent lexicons: precision holds at 1.00 for the capable agent (no planted false cognate taken at any level); what divergence moves is only the weak model's confident-error rate.

Where the negotiation is two-sided



Give the provider its own deliverability to weigh and the second agent diverges from a capacity-blind oracle exactly where paths saturate; the naive oracle offers an undeliverable service in 52% of cells, every reasoning provider none.

When is building the shared reference worth the cognition it costs?

Constructing the reference is itself an act of cognition. Weighing that cost against what it buys - the close reached versus the reasoning spent - turns "always build one?" into a rule.

Load-bearing - where no standard exists

A capable agent with no reference finds only a fifth of the true matches (resolved fraction 0.20); building the reference itself carries it to 0.80, and to 0.90 with a decisive experiment, for a little extra cognition. With no standard to bind through, that construction spend is what buys the close.

Redundant - where an effective reference already exists

In configuration and observability, binding through a given reference reaches the same close for a fraction of the cognition - a strong agent binds through it in a few hundred reasoning tokens against several thousand to construct one. Building what you already have is wasted effort.

Counterproductive - for a weak agent, anywhere

Construction is capability-gated: the weak agent's cost explodes to tens of thousands of reasoning tokens and the close it reaches gets worse, not better. Never hand construction to a weak agent.

Build the reference where there is none and the agent can build a good one; otherwise bind through the one that exists - a standard is simply a constructed reference, amortised across everyone who binds through it.

The lift itself can be done by an agent

WHAT HOLDS

The agent's lift reconciles the same

In the cases, each concept's gloss and example come ready-made, standing for what a reasoning side would say of itself. Give an agent only a side's bare schema (its names, kinds, relations and records) and ask it to write those itself. The capable agent then reconciles its own lift just as it does the ready-made one: identical in all four settings with a shared reference, and close behind without one.

WHAT IT COSTS

A good lift gets dearer as capability falls

The agent writes a gloss and example for every concept. But the effort climbs steeply as the model weakens: a few hundred reasoning tokens for the strong agent, several thousand for the mid one. On the hardest standard-free case the mid agent's own glosses cost it a match the ready-made lift got right. A good lift, like using one, is gated by cognitive power.

...so the results do not rest on the glosses being written in advance

Who writes the lift, a person ahead of time or the agent on the spot, does not change the reconciliation where the study leans on it. What is still assumed is the lift from raw, messy schema, and a cold start before any records exist. Both are natural next steps on real data.

The agent naming its evidence, on invented vocabulary

On the relabelled control, the recognisable names replaced by invented ones so only the structure remains, the strong agent records the surface signals it used and the look-alike it ruled out (its own stated reasoning):

cap-point · terminates a flux-span

“A span termination; ruled out the conduit-cap-point, which originates a conduit rather than terminating a span.” → the point where a span ends at a vertex.

conduit-cap-point · originates a flux-conduit

“A conduit-origin termination; ruled out the cap-point, which terminates spans, and the conduit-circuit-endpoint, an endpoint of the conduit.” → the head from which a conduit originates.

flux-conduit · a service over the fabric

“A network service; ruled out the flux-span, a single link rather than an end-to-end service.” → a service that originates at the conduit head.

The names are invented, so the reading is structural, not recalled; and it is relational, since the conduit leans on the head that originates it. This is the lift's counterpart to the negotiation and the decisive experiment.

A demonstration of how it works, not a statistical survey

Each setting is built from seeded cases, designed to exercise one mechanism and seed the look-alike traps the agents must avoid. To guard against any one case being chosen to work, each setting is exercised on three independently-built cases, and the signatures hold across them (previous slide). The study shows the mechanism works and how it works; it does not estimate how often it would succeed across a whole population of real networks. What remains is now genuinely external.

Larger and more varied cases

Each setting now rests on three constructed cases, but all are authored. More and larger cases, drawn from real networks rather than authored, would show how far these patterns generalise beyond the demonstrations.

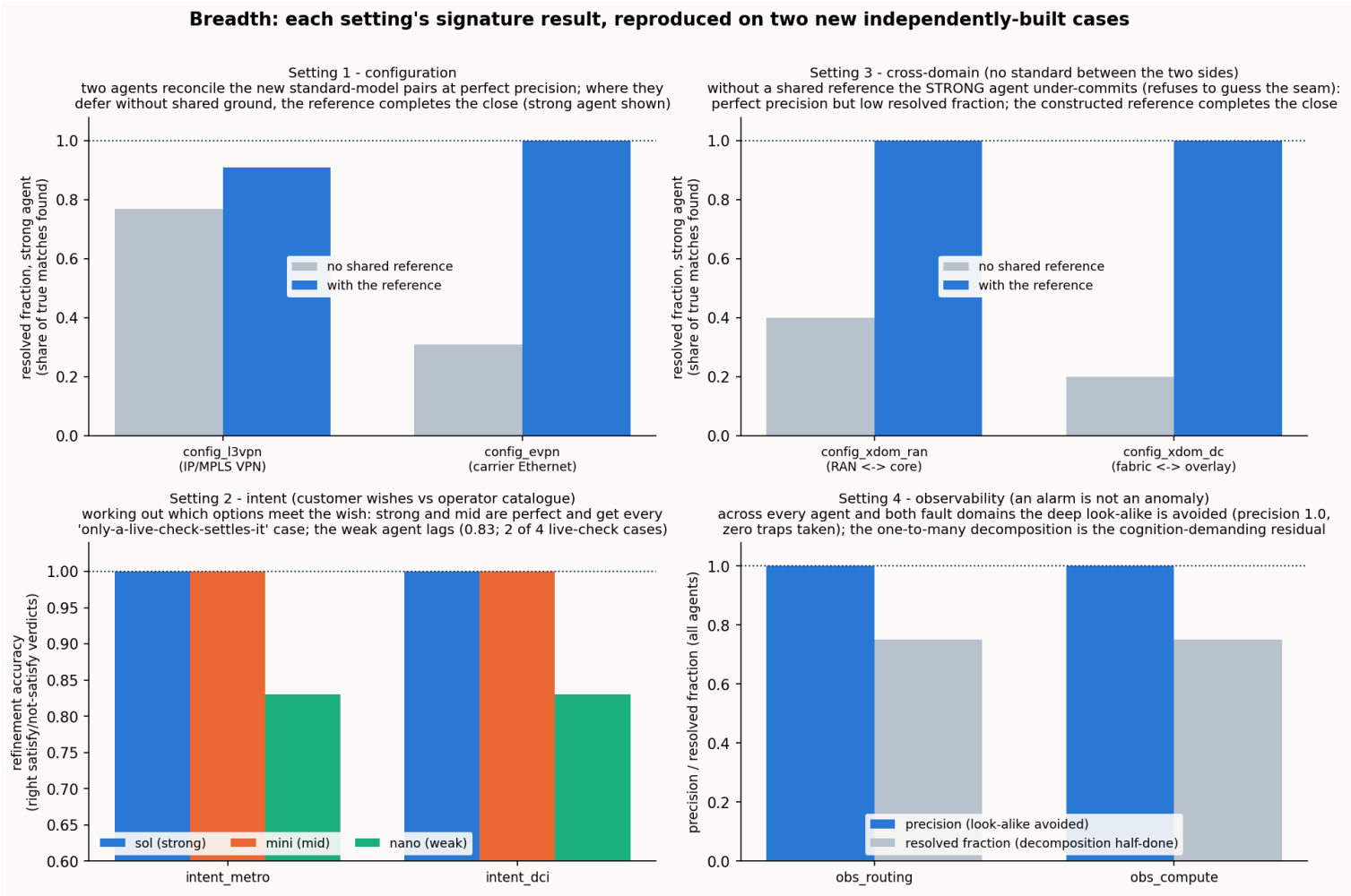
Real operational data

The models, goods and traps are authored, if validated. Grounding them in the genuine divergence between fielded systems is something only carrier collaboration can supply. The same real, messy source is where the remaining lift work lives: building the semantic model from raw schema, and a cold start before any records exist.

Standards-body validation

The reference lexicons and the RFC-anchored modelling choices, stated openly here, would gain from review inside NMOP and the wider standards community.

Every signature, reproduced on new cases



Each setting's signature result on two further independently-built cases. The point is not any single bar but that all four signatures recur on cases built to differ.

IN ONE PARAGRAPH

Cognition completes the reconciliation; the pragmatic layer is where the work now lies

Two systems that can reason do not need a standard agreed in advance to work together. They lift their data into portable, self-describing semantic models and reconcile those ad hoc, for the occasion; the lift that produces them is itself an act an agent can perform, from a bare schema surface at every capability. Across configuration, intent, cross-domain and observability, at the fully-cognitive end the reconciliation completes autonomously in every case, negotiations and significance verdicts included, with no standard and no human.

A thin reference earns its place: it partly substitutes for a capable agent's reasoning and supplies the facts a live agent can no longer get from an inert side. Across many systems, it also makes the reconciliation work grow linearly instead of quadratically. But its reach ends at information; it never provides the authority to decide.

What remains is the pragmatic layer, the one layer no static model holds: it is judged, not read, and resolved live by the operation. Pushing into it, and how capable an agent must be to work there, is where the work now lies.